

Topic P5

Nuclear physics

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, common misconceptions, worked applications and exam-style practice for this topic.

Syllabus map

P5.1	The nuclear model of the atom
P5.2	Radioactivity
P5.3	Fission and fusion

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Atomic structure and nuclide notation

An atom has a tiny positive nucleus containing protons and neutrons, surrounded by electrons. Proton number Z identifies the element; nucleon number A is protons plus neutrons. Isotopes have the same Z but different neutron numbers.

$$\text{neutrons} = A - Z$$

Worked example

For uranium-235, $Z=92$, so neutrons = $235-92 = 143$.

Examiner insight

Changing electrons forms ions; changing neutrons forms isotopes.

2. Radioactive decay and detection

Unstable nuclei decay randomly and spontaneously. Activity is decays per second in becquerels. Count rate includes background, which should be measured and subtracted. A Geiger-Müller tube detects ionising radiation.

Examiner insight

Random means the exact nucleus and time cannot be predicted; spontaneous means unaffected by temperature, pressure or chemical state.

3. Alpha, beta and gamma

Alpha is a helium nucleus: strongly ionising, short range, stopped by paper. Beta-minus is a fast electron: moderate ionisation, stopped by thin aluminium. Gamma is electromagnetic radiation: weakly ionising, highly penetrating, reduced by thick lead or concrete.

Examiner insight

Penetration and ionising ability follow opposite trends.

4. Half-life

Half-life is the time for the number of undecayed nuclei, activity or corrected count rate to halve. Decay is exponential, so the same fraction decays in equal time intervals, not the same number.

$$\text{remaining fraction} = (1/2)^n$$

Worked example

After three half-lives, activity is $(1/2)^3 = 1/8$ of the initial value.

Examiner insight

Subtract background before using count-rate data.

5. Applications and safety

Penetration and ionisation determine use: alpha in smoke detectors, beta for thickness control, gamma for sterilisation and tracers. Reduce exposure using shorter time, greater distance and shielding. Sources are handled with tongs and stored in lead-lined containers.

Examiner insight

Do not describe radiation as remaining inside an irradiated object unless contamination has occurred.

6. Fission and fusion

Fission splits a heavy nucleus after neutron absorption, releasing energy and more neutrons; a controlled chain reaction powers a reactor. Fusion joins light nuclei and powers stars, but requires extremely high temperature and pressure to overcome repulsion.

Examiner insight

Control rods absorb neutrons; the moderator slows neutrons; coolant transfers thermal energy.

Exam technique

High-value habits

Use precise definitions, show every calculation step, include units, apply concepts to unfamiliar situations, and answer the command word directly.

Retrieval checklist

- Recall all definitions and equations without notes.
- Explain one unfamiliar application using a linked chain of reasoning.
- Identify the first error in each incorrect response.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic P5

nuclear physics

Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which particles are in the nucleus? [1]
 - A protons and neutrons
 - B protons and electrons
 - C neutrons and electrons
 - D electrons only

2. Two isotopes of the same element have different numbers of [1]
 - A protons
 - B electrons in neutral atoms
 - C neutrons
 - D nuclear charge

3. What is random about radioactive decay? [1]
 - A the type of element
 - B which nucleus decays and when
 - C the existence of a nucleus
 - D the charge of a proton

4. Which radiation is most ionising? [1]
 - A alpha
 - B beta
 - C gamma
 - D infrared

5. Which material stops beta radiation effectively? [1]
 - A paper only
 - B thin aluminium
 - C air only
 - D a magnetic field

6. Which radiation is electromagnetic? [1]
 - A alpha
 - B beta
 - C gamma
 - D neutron

7. A corrected count rate falls from 800 to 100 counts/min. How many half-lives pass? [1]
- A 2
 - B 3
 - C 4
 - D 8
8. Which action reduces radiation dose? [1]
- A longer exposure
 - B smaller distance
 - C suitable shielding
 - D holding the source by hand
9. What starts nuclear fission? [1]
- A absorption of a neutron
 - B emission of visible light
 - C cooling a nucleus
 - D loss of an electron shell
10. Where does fusion occur naturally? [1]
- A in stars
 - B in cold metals
 - C in batteries
 - D in transformers

Section B · Structured questions

11 A neutral atom has proton number 17 and nucleon number 37.

(a) State the number of protons. [1]

(b) Calculate the number of neutrons. [1]

(c) State the number of electrons. [1]

(d) Explain how another isotope of this element differs. [2]

(e) Describe the charge and approximate relative mass of a proton, neutron and electron. [3]

[Total: 8]

12 A detector records 86 counts/min near a source. Background is 22 counts/min.

(a) Calculate the corrected count rate. [1]

(b) Explain why background must be subtracted. [2]

(c) State two sources of background radiation. [2]

(d) Explain what random and spontaneous mean for nuclear decay. [3]

[Total: 8]

13 A radioactive source emits alpha, beta and gamma radiation.

(a) State the nature of each emission. [3]

(b) Compare their ionising abilities. [2]

(c) State suitable shielding for each. [2]

(d) Identify which is deflected most by an electric field. [1]

[Total: 8]

14 A sample has an initial corrected count rate of 960 counts/min and a half-life of 6.0 h.

(a) Calculate the corrected count rate after 18 h. [3]

(b) Calculate the time for the rate to fall to 60 counts/min. [2]

(c) Explain why count rate never reaches exactly zero in the model. [2]

(d) State one medical use of gamma radiation. [1]

[Total: 8]

15 A nuclear reactor uses fission.

(a) Describe fission of a uranium nucleus. [3]

(b) Explain how a chain reaction occurs. [2]

(c) State the functions of control rods and coolant. [2]

(d) State one difference between fission and fusion.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Award marks for correct physics and method, not only the final value. Mark each part separately.

Solutions · Multiple choice

1 A

2 C

3 B

4 A

5 B

6 C

7 B

8 C

9 A

10 A

Solutions · Structured questions

11. 11(a) 17 [1]. (b) $37-17=20$ [1]. (c) 17 [1]. (d) Same protons but different neutrons [2]. (e) proton +1 and mass 1; neutron 0 and mass 1; electron -1 and very small mass [3].

12. 12(a) $86-22=64$ counts/min [1]. (b) Detector records environmental radiation as well as the source [2]. (c) Any two valid sources [2]. (d) Exact nucleus/time unpredictable; rate unaffected by external conditions [3].

13. 13(a) Alpha helium nucleus; beta fast electron; gamma electromagnetic wave [3]. (b) alpha greatest, beta intermediate, gamma least [2]. (c) paper, aluminium, thick lead/concrete [2]. (d) beta [1].

14. 14(a) Three half-lives: $960 \rightarrow 480 \rightarrow 240 \rightarrow 120$ counts/min [3]. (b) Four half-lives = 24 h [2]. (c) Exponential decay leaves a fraction after every interval [2]. (d) Valid medical use [1].

15. 15(a) Heavy nucleus absorbs neutron and splits, releasing energy and neutrons [3]. (b) Released neutrons cause further fissions [2]. (c) rods absorb neutrons; coolant removes thermal energy [2]. (d) Valid contrast [1].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation choice, rearrangement, unit conversion, graph/diagram interpretation or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.